

GRAZ THALERHOF, Austria

WMO: 112400

Lat: 46.9806N

Lon: 15.4400E

Elev: 363

StdP: 97.04

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-8.9	-14.1	1.2	-7.0	-11.9	1.4	-5.3	9.2	1.2	7.7	2.5	0.8	170	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	31.2	20.6	29.7	20.3	28.0	19.5	22.0	28.0	21.2	27.3	20.4	26.0	3.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	15.5	23.9	19.2	14.6	23.0	18.6	14.1	22.4	66.1	28.1	63.0	27.2	60.2	26.0	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.2	4.9	-14.8	33.9	3.3	2.1	-17.1	35.5	-19.0	36.7	-20.9	37.9	-23.3	39.5		
(5)			-14.9	23.5	3.3	1.0	-17.3	24.2	-19.2	24.8	-21.0	25.3	-23.3	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	-0.8	1.6	6.0	11.0	15.5	19.3	20.9	20.2	15.6	10.8	5.4	0.0	(6)
(7)		DBStd	8.42	3.90	4.33	4.09	3.66	3.32	3.32	2.92	2.90	3.10	3.77	3.96	3.70	(7)
(8)		HDD10.0	1229	335	235	133	32	2	0	0	0	1	37	144	310	(8)
(9)		HDD18.3	3099	593	468	383	221	98	29	11	14	92	234	388	568	(9)
(10)		CDD10.0	1411	0	1	8	61	174	278	337	317	169	61	6	0	(10)
(11)		CDD18.3	241	0	0	0	0	12	57	89	73	10	0	0	0	(11)
(12)		CDH23.3	2256	0	0	0	17	151	525	821	655	82	6	0	0	(12)
(13)		CDH26.7	652	0	0	0	1	26	152	259	206	8	0	0	0	(13)
(14)	Wind	WSAvg	1.7	1.3	1.5	2.0	2.2	2.2	2.0	1.9	1.7	1.5	1.3	1.3	1.2	(14)
(15)	Precipitation	PrecAvg	886	26	35	44	55	90	122	130	124	92	66	58	38	(15)
(16)		PrecMax	1442	66	111	126	133	219	270	276	236	189	216	130	118	(16)
(17)		PrecMin	577	0	0	1	12	34	22	34	48	15	1	1	1	(17)
(18)		PrecStd	163	17	28	28	31	42	49	62	46	42	49	33	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	17.0	20.6	25.3	29.4	32.8	33.8	33.2	27.8	24.1	18.8	11.8	(19)
(20)			MCWB	6.8	9.5	11.2	15.0	18.2	20.6	21.6	21.2	19.5	16.5	13.1	8.4	(20)
(21)		2%	DB	9.1	13.2	18.2	23.0	26.9	30.2	31.2	31.0	25.5	21.2	15.6	8.7	(21)
(22)			MCWB	5.7	7.4	10.3	13.4	17.3	20.6	20.8	20.5	18.1	15.2	11.7	6.1	(22)
(23)		5%	DB	6.8	10.8	15.9	20.9	24.9	28.2	29.3	29.0	23.9	19.2	13.0	6.7	(23)
(24)			MCWB	4.2	6.3	9.3	12.8	16.5	19.7	20.2	20.3	17.4	14.4	10.4	5.0	(24)
(25)		10%	DB	4.8	8.2	13.6	18.9	22.9	26.5	27.9	27.0	22.0	17.0	11.1	5.0	(25)
(26)			MCWB	3.2	4.7	8.2	11.7	15.5	18.7	19.6	19.4	16.2	13.4	9.1	3.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	9.7	12.1	16.0	19.5	22.5	23.2	22.9	20.4	17.4	13.8	9.2	(27)
(28)			MCDB	10.8	15.9	18.4	23.2	25.9	29.2	29.8	29.2	25.4	21.7	17.2	11.4	(28)
(29)		2%	WB	6.1	8.0	10.8	14.6	18.3	21.4	22.2	21.7	19.2	16.3	12.3	6.8	(29)
(30)			MCDB	8.2	12.3	16.5	20.7	24.4	27.7	28.0	28.0	23.8	19.8	14.7	7.9	(30)
(31)		5%	WB	4.6	6.7	9.8	13.4	17.3	20.5	21.2	21.0	18.2	15.2	10.9	5.1	(31)
(32)			MCDB	6.2	9.9	14.9	18.9	23.2	26.5	27.2	27.0	22.3	18.0	12.9	6.0	(32)
(33)		10%	WB	3.1	5.2	8.8	12.4	16.3	19.6	20.3	20.1	17.3	14.1	9.6	3.9	(33)
(34)			MCDB	4.3	7.6	12.9	17.5	21.5	25.0	26.0	25.4	20.7	16.7	10.9	4.7	(34)
(35)	Mean Daily Temperature Range	MDBR	7.4	9.5	11.0	11.5	11.3	11.1	11.3	11.2	10.3	9.8	7.2	6.7	(35)	
(36)		5% DB	MCDBR	11.3	14.1	15.7	15.8	14.7	14.1	13.9	14.4	12.9	13.1	11.4	9.9	(36)
(37)			MCWBR	7.7	8.5	8.4	7.2	5.9	5.3	5.0	5.3	6.1	7.3	7.2	7.3	(37)
(38)			MCDBR	10.3	13.0	14.3	13.7	13.1	13.0	12.7	12.9	11.6	11.4	9.9	8.6	(38)
(39)		5% WB	MCWBR	7.4	8.2	7.9	6.7	5.9	5.3	5.2	5.5	5.9	6.7	6.6	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.304	0.331	0.370	0.416	0.418	0.424	0.426	0.426	0.397	0.370	0.337	0.304	(40)	
(41)		taud	2.433	2.374	2.304	2.201	2.245	2.268	2.261	2.273	2.327	2.380	2.430	2.472	(41)	
(42)		Ebn at Noon	786	833	847	837	849	843	835	818	809	774	734	745	(42)	
(43)		Edh at Noon	68	88	110	134	133	131	130	124	107	88	69	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.24	2.09	3.18	4.18	4.97	5.50	5.50	4.87	3.50	2.35	1.30	1.02	(44)	
(45)		RadStd	0.15	0.29	0.39	0.46	0.52	0.39	0.43	0.54	0.45	0.25	0.20	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (97 neighbors)	+0.52	N/A	N/A	+0.87	+0.48	+0.40	-102	-140	+90	+57

Nomenclature: See separate page